

Overview of Speech recognition

Subject: Speech recognition

1.

Helicopters The problems of achieving high recognition accuracy under stress and noise are particularly relevant in the helicopter environment as well as in the fighter environment. The acoustic noise problem is actually more severe in the helicopter environment, because of the high noise levels, and because helicopter pilots, in general, do not wear a facemask, which would reduce acoustic noise in the microphone. Substantial test and evaluation programmes, notably by the U.S. Army Avionics Research and Development Activity (AVRADA) and by the Royal Aerospace Establishment (RAE) in the UK. Work in France included speech recognition in the Puma helicopter. Voice applications include control of communication radios, navigation systems, and an automated target handover system. The overriding issue for voice is the impact on pilot effectiveness. Encouraging results are reported for the AVRADA tests, although these represent only a feasibility demonstration in a test environment. Much remains to be done both in speech recognition and in overall speech technology in order to consistently achieve performance improvements in operational settings.

2.

Speech recognition (automatic speech recognition (ASR), computer speech recognition, or speech-to-text (STT)) is a sub-field of computational linguistics concerned with methods and technologies that translate spoken language into text or other interpretable forms. Speech recognition applications include voice user interfaces, where the user speaks to a device, which "listens" and processes the audio. Common voice applications include interpreting commands for calling, call routing, home automation, and aircraft control. These applications are called direct voice input. Productivity applications include searching audio recordings, creating transcripts, and dictation. Speech recognition can be used to analyse speaker characteristics, such as identifying native language using pronunciation assessment. Voice recognition (speaker identification) refers to identifying the speaker, rather than speech contents. Recognizing the speaker can simplify the task of translating speech in systems trained on a specific person's voice. It can also be used to authenticate the speaker as part of a security process.

3.

Security Speech recognition can become a means of attack, theft, or accidental operation. For example, activation words like "Alexa" spoken in an audio or video broadcast can cause devices in homes and offices to start listening for input inappropriately, or possibly take an unwanted action. Voice-controlled devices may be accessible to unauthorized users. Attackers may be able to gain access to personal information, like calendars, address book contents, private messages, and documents. They may also be able to impersonate the user to send messages or make online purchases. Two attacks have been demonstrated that use artificial sounds. One transmits ultrasound and attempts to send commands without people noticing. The other adds small, inaudible distortions to other speech or music that are specially crafted to confuse the specific speech recognition system into recognizing music as speech, or to make what sounds like one command to a human sound like a different command to the system.

4.

2010s By early the 2010s, speech recognition was differentiated from speaker recognition, and speaker independence was considered a major breakthrough. Until then, systems required a "training" period for each voice. In 2017, Microsoft researchers reached the human parity milestone of transcribing conversational speech on the widely benchmarked Switchboard task. Multiple deep learning models were used to improve accuracy. The error rate was reported to be as low as 4 professional human transcribers working together on the same benchmark.

5.

1987 – The back-off model enabled language models to use multiple-length n-grams, and CSELT used HMM to recognize languages (in software and hardware, e.g. RIPAC). At the end of the DARPA program in 1976, the best computer available to researchers was the PDP-10 with 4 MB of RAM. It could take up to 100 minutes to decode 30 seconds of speech. Practical products included:

6.

People with disabilities Speech recognition programs can provide many benefit to those with disabilities. For individuals who are deaf or hard of hearing, speech recognition software can be used to generate captions of conversations. Additionally, individuals who are blind (see blindness and education) or have poor vision can benefit from listening to textual content, as well as garner more functionality from a computer by issuing commands with their voice. The use of voice recognition software, in conjunction with a digital audio recorder and a personal computer running word-processing software, has proven useful for restoring damaged short-term memory capacity in individuals who have suffered a stroke or have undergone a craniotomy. Speech recognition has proven very useful for those who have difficulty using their hands due to causes ranging from mild repetitive stress injuries to disabilities that preclude the use of conventional computer input devices. Individuals with physical disabilities can use voice commands and transcription to navigate electronics hands-free. In fact, people who developed RSI from keyboard use became an early and urgent market for speech recognition. Speech recognition is used in deaf telephony, such as voicemail to text, relay services, and captioned telephone. Individuals with learning disabilities who struggle with thought-to-paper communication may benefit from the software, but the product's fallibility remains a significant consideration for many. In addition, speech to text technology is only an effective aid for those with intellectual disabilities if the proper training and resources are provided (e.g. in the classroom setting). This type of technology can help those with dyslexia, but the potential benefits regarding other disabilities are still in question. Mistakes made by the software hinder its effectiveness, since misheard words take more time to fix.

7.

Speech recognition systems are based on HMMs. These are statistical models that output a sequence of symbols or quantities. HMMs are used in speech recognition because a speech signal can be viewed as a piecewise stationary signal or a short-time stationary signal. In a short time scale (e.g. 10 milliseconds), speech can be approximated as a stationary process. Speech can be thought of as a Markov model for many stochastic purposes. HMMs are popular because they can be trained automatically and are simple and computationally feasible. An HMM outputs a sequence of n-dimensional real-valued vectors (where n is an integer such as 10), outputting one every 10 milliseconds. The vectors consist of cepstral coefficients, obtained by a Fourier transform of a short

window of speech and decorrelating the spectrum using a cosine transform, then taking the first (most significant) coefficients. The HMM tends to have, in each state, a statistical distribution that is a mixture of diagonal covariance Gaussians, which give a likelihood for each observed vector. Each word, or (for more general speech recognition systems), each phoneme, has a different output distribution; an HMM for a sequence of words or phonemes is made by concatenating the individual trained HMMs for the separate words and phonemes. Speech recognition systems use combinations of standard techniques to improve results. A typical large-vocabulary system applies context dependency for the phonemes (so that phonemes with different left and right context have different realizations as HMM states). It uses cepstral normalization to handle speaker and recording conditions. It might use vocal tract length normalization (VTLN) for male-female normalization and maximum likelihood linear regression (MLLR) for more general adaptation. The features use delta and delta-delta coefficients to capture speech dynamics, and in addition might use heteroscedastic linear discriminant analysis (HLDA); or might use splicing and LDA-based projection, followed by HLDA or a global semi-tied covariance transform (also known as maximum likelihood linear transform (MLLT)). Many systems use discriminative training techniques that dispense with a purely statistical approach to HMM parameter estimation and instead optimize some classification-related measure of the training data. Examples are maximum mutual information (MMI), minimum classification error (MCE), and minimum phone error (MPE).